

# AHAAN KOTIAN

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## Publications

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- MAMMOTH: A Multi-Modal End-to-End Policy for Off-Road Mobility Robust to Missing Modality (Accepted - IROS '26) – **Ahaan Kotian**, Shivani Subramanyan, Suresh Sundaram ([project page](#))
- MS-DOTT: A Comprehensive Multi-Modal Dataset for Off-Road Traversability Estimation (Under Submission - RAAI '26) – Shivani Subramanyan, Saksham Sharma, Samahit SA, **Ahaan Kotian**, Suresh Sundaram ([video](#))
- A Simulation Framework for Surgical Subtask Automation (Accepted - RCAAI 2024) – **Ahaan Kotian**, Rugved Palodkar, Vaibhav Narwane
- Generative Deep Learning for Drug Discovery (Accepted - ICTEAH 2025) – Rugved Palodkar, **Ahaan Kotian**, Prasanna Shete

## Work Experience

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### Indian Institute of Science

June 2025 – Ongoing

*Pre-Doctoral Researcher*

*Bengaluru, Karnataka*

- Developed MAMMOTH, a multi-modal end-to-end policy for off-road navigation capable of generalizing to missing modalities and diverse, challenging lighting conditions. (Accepted to IROS'26 as first-author)
- Developed MS-DOTT, a large-scale multi-modal off-road dataset for traversability estimation in varied lighting conditions. (Submitted to RAAI'26)
- Developed a novel positional encoding technique for transformer-based spatio-temporal multi-modal fusion for asynchronous sensor streams. (Under development for first-author A\* publication)
- Designing VLM-guided asynchronous end-to-end navigation policies and fusion techniques for outdoor, in-the-wild navigation.
- Engineered a robust navigation policy which takes inputs RGB, Thermal, 3D pointclouds, and ego-velocity observations and generates physically-aware trajectories in ego frame robust to lighting conditions and missing modality scenarios.

### Indian Institute of Science

January 2025 – June 2025

*Research Intern*

*Bengaluru, Karnataka*

- Executed the implementation of multiple end-to-end learning models successfully, from high-fidelity CARLA simulations to physical hardware platforms
- Developed a full navigation stack for a custom All-Terrain Vehicle (ATV) using Nav2 and 3D LiDAR for autonomous path planning
- Programmed a custom behavior tree using the BehaviorTree.CPP library to manage complex navigation logic for an Ackermann-steered vehicle in Gazebo.

### Indian Institute of Technology, Bombay

May 2024 – July 2024

*Research Intern*

*Mumbai, Maharashtra*

- Researched and implemented a planning framework for mobile manipulators focused on complex door opening and traversal tasks.
- Engineered a custom global planner in C++ utilizing ARA\* search, custom state space, and cost formulation within the ROS navigation stack.
- Utilized MoveIt! and Gazebo for simulation and testing of the mobile manipulator's interaction with environmental objects.

### Orion Racing India

Dec 2021 – Sep 2024

*Driverless System Lead*

*Mumbai, Maharashtra*

- Leading a technical team of 10+ people to develop formula student driverless car prototype.
- Designed Dijkstras, A\*, RRT, RRT\* global planning algorithms for 1:10 scale autonomous racing car.

- Built and simulated robust PD and Pure Pursuit algorithms using python3 and ROS.
- Collaborated with a team to integrate the software pipeline using ROS.
- Executing training programs for new recruits to equip them with essential technical skills and knowledge.
- Handling sponsorships and partnerships, contributing to a robust financial foundation for the team's initiatives.

## Education

**K.J. Somaiya School of Engineering, Mumbai**

**2021 – 2025**

*BTech in Computer Engineering and Minor in Robotics & AI*

*CGPA: 9.60/10; 10/10(Minor)*

## Projects

### Visual Question Answering for Autonomous Driving 🤖

- Implemented a Visual Question Answering (VQA) pipeline to evaluate the scene understanding capabilities of diverse Vision-Language Models (VLMs) in autonomous driving contexts.
- Curated and pre-processed the DriveLM dataset, comprising RGB observations paired with logic-based questions and answer labels about perception, planning, prediction and behaviour.
- Fine-tuned SmolVLM and Qwen2.5-VL-3B-Instruct models using Parameter-Efficient Fine-Tuning (PEFT) techniques, specifically LoRA and QLoRA.
- Attained exact-match accuracy scores of 56.68% with SmolVLM and 65.78% with Qwen2.5-VL-3B-Instruct on unseen test data.
- Technology Used: PyTorch, HuggingFace, python3.

### Generative Deep Learning for Drug Discovery 🤖

- Generation of novel chemical compounds for drug discovery for inhibiting progression of Alzheimer's disease using deep learning.
- Sourced generic and focused chemical dataset from ChEMBL in SMILES syntax to leverage transfer learning.
- Designed custom tokeniser to process chemical syntax including double-character elements and heteroatoms.
- Achieved accuracy of 82.68% on training data and 83.50% on validation data using an Improved LSTM model.
- Generated 4 valid molecules and utilised RDKit for checking validity and generation of their 2D structure.
- Technology Used: PyTorch, RDKit, ChEMBL, python3.

### Artificial Potential Field Planner for Multi-Robot Simulation 🤖

- Simulated multi-robot coordination in Gazebo using an Artificial Potential Field (APF) planner to achieve arbitrary formations around a target pose.
- Architected the simulation framework and optimized URDF models for the efficient execution of multiple Volta robots in Gazebo for low-resource environments.
- Technology Used: python3, ROS, Gazebo, Rviz.

## Technical Skills

**Languages:** Python, C/C++, SQL, JAVA, MATLAB, HTML, CSS, JavaScript, LaTeX

**Frameworks & Library:** PyTorch, Tensorflow, Hugging Face, OpenCV, Scikit-learn, NumPy, Matplotlib, ROS/ROS2, Moveit!, Gazebo, BehaviourTreeCPP

**Environment/Tools:** Docker, SLURM, Git/GitHub, Linux, CUDA, Anaconda, Google Colab, Jupyter, Weights & Biases

## Leadership / Extracurricular

- Conducted a FDP (Faculty Development Programme) session at IISc on ROS2 and Nav2 which included hands-on tutorials and exercises.
- Served on the organizing committee of the International Conference on Emerging Technology in Autonomous Aerial Vehicles (ETA AV), contributing to overall event coordination.
- Managed paper presentation scheduling, prepared slide decks, conducted plagiarism checks, and supported the Q&A and session-moderation team.